

Building a Model Card

What Is a Model Card and Why Is It Important?

A model card is a document (typically the README .md file) that is just as important as the model and tokenizer files in a model repository.

It serves to:

- Provide central documentation for the model
- Make the model reusable by others
- Enable reproducibility of results
- Clearly state limitations, biases, and applicable contexts
- Help others build artifacts on top of the model

In short, publishing a model without a model card means omitting essential context.

Why Is It Important to Document Training and Evaluation?

Documenting the training and evaluation process helps:

- Others understand what level of performance to expect
- Identify which dataset(s) were used
- Clarify preprocessing and postprocessing steps
- Detect limitations and potential biases
- Define where the model is appropriate — and where it is not

This transparency is critical for responsible and reliable model usage.

Where Is the Model Card Written?

The model card is created in:

README.md

This file is written in Markdown format.

Origin of the Model Card Concept

The concept of the “Model Card” originated from Google research:

“Model Cards for Model Reporting” – Margaret Mitchell et al.

This framework emphasizes reproducibility, reusability, and fairness in machine learning systems.

Typical Structure of a Model Card

A well-written model card usually contains the following sections:

1. Model description
2. Intended uses & limitations
3. How to use
4. Limitations and bias
5. Training data
6. Training procedure
7. Evaluation metrics
8. Evaluation results

Below is an explanation of each section.

1) Model Description

This section provides basic information about the model, including:

- Architecture (e.g., BERT, GPT-2, DistilBERT)
- Version
- Whether it was introduced in a research paper
- Availability of an original implementation
- Author
- Copyright / License
- General training details
- Important disclaimers

This is essentially the model's identity section.

2) Intended Uses & Limitations

This section should describe:

- The tasks the model is designed for
- Supported languages, domains, or fields
- Situations where it should not be used
- Scenarios where performance may degrade

Clearly defining scope helps prevent misuse.

3) How to Use

Provide practical usage examples, such as:

- Using the `pipeline()` function
- Using model and tokenizer classes directly
- Sample code snippets

This section helps users get started quickly.

4) Limitations and Bias

Here you should describe:

- Potential data-driven biases
- Demographic imbalances
- Domain limitations
- Ethical concerns

This is important for responsible AI publication.

5) Training Data

Specify:

- Which dataset(s) were used
 - A brief description of each dataset
 - Whether the datasets are public or private
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6) Training Procedure

For reproducibility, include:

- Number of training epochs
- Batch size
- Learning rate
- Optimizer
- Preprocessing steps
- Postprocessing steps

The more transparent this section is, the more trustworthy the model becomes.

7) Metrics (Variables and Metrics)

Clearly state:

- Which evaluation metrics were used (e.g., Accuracy, F1, BLEU, ROUGE)
- Which dataset split (train/validation/test)
- On which dataset evaluation was performed

This makes it easier to compare your model with others.

8) Evaluation Results

Provide:

- The model's performance on evaluation data
 - The decision threshold used (if applicable)
 - Performance across different thresholds (if relevant)
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Examples of Well-Crafted Model Cards

You can review the following models on the Hugging Face Hub:

- `bert-base-cased`
- `gpt2`
- `distilbert`

These provide strong examples of structured and informative model cards.

Model Card Metadata (Header Section)

At the top of the model card, you can include metadata.

The Hugging Face Hub reads this metadata to categorize the model.

Example:

language: fr

license: mit

datasets:

- oscar

This indicates:

- The model is for the French language
- It uses the MIT license
- It was trained on the OSCAR dataset

What Can Be Defined Through Metadata?

- Language
- License
- Tags
- Dataset
- Metrics
- Evaluation results

The Hub uses this metadata for filtering and categorization.